

Periodic Classification of Elements

1. Classification of elements is based on

- A) Colour
- B) Properties
- C) Taste
- D) Size

Answer: B

2. The first attempt to classify elements was made by

- A) Newton
- B) Dobereiner
- C) Dalton
- D) Bohr

Answer: B

3. Dobereiner classified elements into

- A) Groups
- B) Periods
- C) Triads
- D) Series

Answer: C

4. A Dobereiner triad consists of

- A) Two elements
- B) Three elements
- C) Four elements
- D) Five elements

Answer: B

5. In Dobereiner triads the atomic mass of the middle element is approximately

- A) Average of first and third
- B) Sum of others
- C) Difference of others
- D) Half of others

Answer: A

6. Example of Dobereiner triad is

- A) Li, Na, K
- B) C, N, O
- C) H, He, Li
- D) Fe, Co, Ni

Answer: A

7. Law of octaves was proposed by

- A) Mendeleev
- B) Newlands
- C) Moseley
- D) Dalton

Answer: B

8. According to Newlands law of octaves every ____ element had similar properties

- A) Sixth
- B) Seventh
- C) Eighth
- D) Ninth

Answer: C

9. Newlands arranged elements in order of increasing

- A) Atomic mass
- B) Density
- C) Volume
- D) Charge

Answer: A

10. The scientist who proposed the periodic table was

- A) Mendeleev
- B) Rutherford
- C) Thomson
- D) Bohr

Answer: A

11. Mendeleev arranged elements based on

- A) Atomic mass

- B) Atomic number
- C) Density
- D) Colour

Answer: A

12. Mendeleev's periodic law states that properties of elements are periodic functions of their

- A) Atomic number
- B) Atomic mass
- C) Density
- D) Volume

Answer: B

13. Mendeleev left gaps in his periodic table for

- A) Unknown elements
- B) Metals
- C) Gases
- D) Liquids

Answer: A

14. The modern periodic law is based on

- A) Atomic mass
- B) Atomic number
- C) Density
- D) Volume

Answer: B

15. Modern periodic law was proposed by

- A) Moseley
- B) Dalton
- C) Newton
- D) Einstein

Answer: A

16. Modern periodic law states that properties of elements are periodic functions of their

- A) Atomic mass

B) Atomic number

C) Density

D) Volume

Answer: B

17. Horizontal rows in periodic table are called

A) Groups

B) Periods

C) Blocks

D) Series

Answer: B

18. Vertical columns in periodic table are called

A) Periods

B) Groups

C) Blocks

D) Series

Answer: B

19. Total number of periods in periodic table is

A) 5

B) 6

C) 7

D) 8

Answer: C

20. Total number of groups in periodic table is

A) 16

B) 17

C) 18

D) 20

Answer: C

21. Elements in the same group have

A) Similar chemical properties

B) Same mass

C) Same size

D) Same density

Answer: A

22. Atomic number represents number of

A) Neutrons

B) Protons

C) Electrons and neutrons

D) Protons and neutrons

Answer: B

23. Elements are arranged in modern periodic table according to increasing

A) Atomic mass

B) Atomic number

C) Density

D) Volume

Answer: B

24. Elements in a period have same number of

A) Shells

B) Electrons

C) Protons

D) Neutrons

Answer: A

25. Elements in same group have same number of

A) Shells

B) Valence electrons

C) Protons

D) Neutrons

Answer: B

26. Alkali metals belong to group

A) 1

B) 2

C) 17

D) 18

Answer: A

27. Alkaline earth metals belong to group

- A) 1
- B) 2
- C) 16
- D) 17

Answer: B

28. Halogens belong to group

- A) 16
- B) 17
- C) 18
- D) 15

Answer: B

29. Noble gases belong to group

- A) 16
- B) 17
- C) 18
- D) 14

Answer: C

30. Noble gases are

- A) Highly reactive
- B) Inert
- C) Liquids
- D) Metals

Answer: B

31. Sodium belongs to

- A) Alkali metals
- B) Halogens
- C) Noble gases
- D) Alkaline earth metals

Answer: A

32. Chlorine belongs to

- A) Alkali metals

- B) Halogens
- C) Noble gases
- D) Transition metals

Answer: B

33. Helium belongs to

- A) Alkali metals
- B) Noble gases
- C) Halogens
- D) Metalloids

Answer: B

34. Atomic size increases

- A) Across period
- B) Down the group
- C) Both directions
- D) None

Answer: B

35. Atomic size decreases

- A) Down group
- B) Across period
- C) Both
- D) None

Answer: B

36. Metallic character increases

- A) Across period
- B) Down group
- C) Up group
- D) Left to right

Answer: B

37. Non-metallic character increases

- A) Down group
- B) Across period
- C) Up group

D) None

Answer: B

38. Elements with properties of both metals and non-metals are

A) Metals

B) Metalloids

C) Halogens

D) Gases

Answer: B

39. Example of metalloid

A) Silicon

B) Sodium

C) Chlorine

D) Helium

Answer: A

40. Transition elements are located in

A) s block

B) p block

C) d block

D) f block

Answer: C

41. Lanthanides belong to

A) f block

B) s block

C) p block

D) d block

Answer: A

42. Actinides belong to

A) f block

B) p block

C) s block

D) d block

Answer: A

43. The periodic table is divided into

- A) Metals, non-metals, metalloids
- B) Only metals
- C) Only non-metals
- D) Only gases

Answer: A

44. Metals are generally

- A) Good conductors
- B) Poor conductors
- C) Insulators
- D) Gases

Answer: A

45. Non-metals are generally

- A) Conductors
- B) Insulators
- C) Metals
- D) Liquids

Answer: B

46. Metals are usually

- A) Malleable
- B) Brittle
- C) Dull
- D) Soft gases

Answer: A

47. Metals are

- A) Ductile
- B) Non-ductile
- C) Liquid
- D) Invisible

Answer: A

48. Non-metals are generally

- A) Brittle

- B) Malleable
- C) Ductile
- D) Conductors

Answer: A

49. The element with atomic number 1 is

- A) Hydrogen
- B) Helium
- C) Lithium
- D) Oxygen

Answer: A

50. Hydrogen is placed in group

- A) 1
- B) 2
- C) 17
- D) 18

Answer: A

51. Elements in group 17 are

- A) Halogens
- B) Noble gases
- C) Alkali metals
- D) Transition metals

Answer: A

52. Elements in group 18 are

- A) Noble gases
- B) Alkali metals
- C) Halogens
- D) Metals

Answer: A

53. Atomic number increases from

- A) Right to left
- B) Left to right
- C) Top to bottom only

D) Bottom to top

Answer: B

54. The first period contains

A) 2 elements

B) 8 elements

C) 18 elements

D) 10 elements

Answer: A

55. The second period contains

A) 2 elements

B) 8 elements

C) 18 elements

D) 32 elements

Answer: B

56. Elements in the same group show

A) Similar valency

B) Same atomic mass

C) Same size

D) Same density

Answer: A

57. Valency of group 1 elements is

A) 1

B) 2

C) 3

D) 4

Answer: A

58. Valency of group 17 elements is

A) 1

B) 2

C) 3

D) 7

Answer: A

59. Noble gases have valency

- A) 1
- B) 2
- C) 0
- D) 3

Answer: C

60. Periodic table helps to

- A) Predict properties
- B) Identify elements
- C) Study trends
- D) All of these

Answer: D

61. Metals are located on

- A) Left side
- B) Right side
- C) Middle only
- D) Bottom only

Answer: A

62. Non-metals are located on

- A) Left side
- B) Right side
- C) Middle
- D) Bottom

Answer: B

63. Metalloids lie between

- A) Metals and non-metals
- B) Metals only
- C) Non-metals only
- D) Gases only

Answer: A

64. Atomic radius decreases across a period due to

- A) Increased nuclear charge

- B) Decreased electrons
- C) Increased neutrons
- D) Lower charge

Answer: A

65. Ionization energy generally increases

- A) Down group
- B) Across period
- C) Both
- D) None

Answer: B

66. Ionization energy decreases

- A) Down group
- B) Across period
- C) Same everywhere
- D) None

Answer: A

67. Electronegativity increases

- A) Across period
- B) Down group
- C) Same
- D) None

Answer: A

68. Electronegativity decreases

- A) Down group
- B) Across period
- C) Same
- D) None

Answer: A

69. The periodic table helps in studying

- A) Chemical properties
- B) Physical properties
- C) Trends

D) All of these

Answer: D

70. Elements with similar properties are placed in

A) Same group

B) Same period

C) Same block

D) Same row

Answer: A

71. Periodic classification helps to

A) Simplify study of elements

B) Complicate study

C) Remove elements

D) Change properties

Answer: A

72. Metals lose electrons to form

A) Cations

B) Anions

C) Molecules

D) Neutrons

Answer: A

73. Non-metals gain electrons to form

A) Cations

B) Anions

C) Neutrons

D) Atoms

Answer: B

74. Sodium forms

A) Na^+

B) Na^-

C) Na^{2+}

D) Na^{2-}

Answer: A

75. Chlorine forms

- A) Cl^+
- B) Cl^-
- C) Cl^{2+}
- D) Cl^{2-}

Answer: B

76. Metals form

- A) Positive ions
- B) Negative ions
- C) Neutral ions
- D) Molecules only

Answer: A

77. Non-metals form

- A) Positive ions
- B) Negative ions
- C) Neutral atoms
- D) Protons

Answer: B

78. Periodic table was improved by

- A) Moseley
- B) Newton
- C) Dalton
- D) Faraday

Answer: A

79. Atomic number determines

- A) Position of element
- B) Colour
- C) Taste
- D) Density

Answer: A

80. Hydrogen shows properties of

- A) Alkali metals

B) Halogens

C) Both

D) None

Answer: C

81. Periodic table predicts

A) New elements

B) Properties

C) Trends

D) All of these

Answer: D

82. Atomic mass is sum of

A) Protons and neutrons

B) Electrons and neutrons

C) Protons only

D) Neutrons only

Answer: A

83. Electron shells increase

A) Across period

B) Down group

C) Same

D) None

Answer: B

84. Chemical properties depend on

A) Valence electrons

B) Atomic mass

C) Density

D) Colour

Answer: A

85. Group number indicates

A) Valence electrons

B) Shell number

C) Atomic mass

D) Density

Answer: A

86. Period number indicates

A) Shell number

B) Valence electrons

C) Density

D) Charge

Answer: A

87. Alkali metals are

A) Highly reactive

B) Inert

C) Non-reactive

D) Noble gases

Answer: A

88. Halogens are

A) Highly reactive non-metals

B) Metals

C) Noble gases

D) Inert

Answer: A

89. Noble gases are

A) Stable

B) Highly reactive

C) Metals

D) Liquids

Answer: A

90. Helium has

A) 2 electrons

B) 4 electrons

C) 8 electrons

D) 1 electron

Answer: A

91. Neon has

- A) 8 electrons in outer shell
- B) 2 electrons
- C) 1 electron
- D) 4 electrons

Answer: A

92. Elements in same group have

- A) Similar valency
- B) Same atomic mass
- C) Same colour
- D) Same density

Answer: A

93. Periodic trends help to

- A) Predict reactions
- B) Predict properties
- C) Study elements
- D) All of these

Answer: D

94. Metals are mostly

- A) Solid
- B) Liquid
- C) Gas
- D) Plasma

Answer: A

95. Mercury is

- A) Liquid metal
- B) Gas
- C) Solid metal
- D) Non-metal

Answer: A

96. Bromine is

- A) Liquid non-metal

- B) Solid metal
- C) Gas metal
- D) Solid non-metal

Answer: A

97. Periodic classification groups elements by

- A) Similar properties
- B) Similar colour
- C) Similar mass only
- D) Similar density

Answer: A

98. Atomic number of carbon is

- A) 6
- B) 8
- C) 12
- D) 14

Answer: A

99. Atomic number of oxygen is

- A) 6
- B) 7
- C) 8
- D) 9

Answer: C

100. Periodic table is useful for

- A) Studying elements systematically
- B) Ignoring elements
- C) Destroying elements
- D) Changing atoms

Answer: A